

## CLAIMS

### What is claimed is:

1. A computer-implemented method for determining an entropic response profile of a material, the method comprising:
  - receiving, at a processor, input data comprising one or more fundamental atomic properties of the material, said properties including at least an atomic mass and a chemical family classification;
  - processing, by the processor, the input data using a predictive engine, wherein the predictive engine is trained on experimental data from ultracold plasma expansion tests; and
  - generating, by the processor, an output entropic response profile for the material, the profile comprising a predicted entropic relaxation time ( $\tau$ ).
2. The method of claim 1, wherein the output entropic response profile further comprises a predicted entropic buffer time ( $t_{break}$ ).
3. The method of claim 1, wherein the predictive engine is configured to determine the predicted entropic relaxation time ( $\tau$ ) by applying a baseline scaling law, wherein said baseline scaling law defines a linear relationship between the atomic mass and the entropic relaxation time based on data from noble gas elements.
4. The method of claim 3, wherein the predictive engine is further configured to apply a quantifiable entropic anomaly offset to the baseline scaling law, wherein the anomaly offset is determined based on the chemical family classification of the material.
5. The method of claim 4, wherein the chemical family classification is an alkali metal, and the entropic anomaly offset corresponds to an accelerated relaxation, resulting in a predicted entropic relaxation time ( $\tau$ ) that is shorter than predicted by the baseline scaling law.

6. The method of claim 4, wherein the chemical family classification is a halogen, and the entropic anomaly offset corresponds to a two-stage buffered relaxation, wherein the output entropic response profile comprises a non-zero entropic buffer time ( $t_{break}$ ) and a predicted entropic relaxation time ( $\tau$ ) that is longer than predicted by the baseline scaling law.
7. A system for determining an entropic response profile of a material, the system comprising:
  - a data input interface configured to receive fundamental atomic properties of the material, said properties including at least an atomic mass and a chemical family classification;
  - a processor; and
  - a memory storing instructions that, when executed by the processor, cause the system to perform the method of any one of claims 1 to 6.
8. The method of claim 1, further comprising the step of selecting or designing a material for a specific application based on its generated entropic response profile, wherein said application relates to one of controlled energy release or controlled energy storage.